Customer Churn Prediction in the Banking Sector

(Analysis and Predictive Modelling Report)

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1. Executive Summary

This report documents an end-to-end churn analysis built on a 10,000-row retail banking dataset. Roughly one in five customers (20.4%) has churned historically, and the goal of the project is to understand why, and to flag which currently active customers are most at risk so retention teams can act early.

Three models were compared: Logistic Regression, Random Forest, and XGBoost. XGBoost gave the strongest overall discrimination (ROC-AUC 0.867), narrowly ahead of Random Forest (0.855), with Logistic Regression well behind on ranking power (0.777) but the most interpretable of the three. Across every model, the same handful of signals kept surfacing as the real churn drivers: customer age, whether the customer is an active member, how many products they hold, their balance, and whether they are based in Germany.

The back half of this report turns those findings into something operational: a risk-tiering scheme (Low / Medium / High) and a set of retention actions mapped to each tier, both of which now also power an interactive churn-prediction dashboard built alongside this report.

2. Objective & Scope

- Understand the historical drivers of customer churn using exploratory data analysis.
- Build and compare multiple classification models to predict churn probability for a customer.
- Identify the features that matter most, and where the linear (Logistic Regression) and non-linear (XGBoost) views of importance agree or disagree.
- Translate model output into a risk-tiering system and concrete retention strategies.
- Package the model as an interactive dashboard for front-line use.

3. Dataset Overview

The dataset contains 10,000 customers of a retail bank operating in France, Germany, and Spain. After dropping identifier-only columns (RowNumber, Surname), the working feature set was:

- CreditScore, Geography, Gender, Age, Tenure (years with bank)
- Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary
- Exited — the target (1 = churned, 0 = retained)

Data quality was clean going in: no missing values and no duplicate rows were found, and Gender was encoded numerically (Female = 0, Male = 1) for modeling. The target is imbalanced — 7,963 retained vs 2,037 churned — which is why class_weight='balanced' was used for Logistic Regression and Random Forest, and why recall/F1/ROC-AUC are reported alongside accuracy rather than accuracy alone.

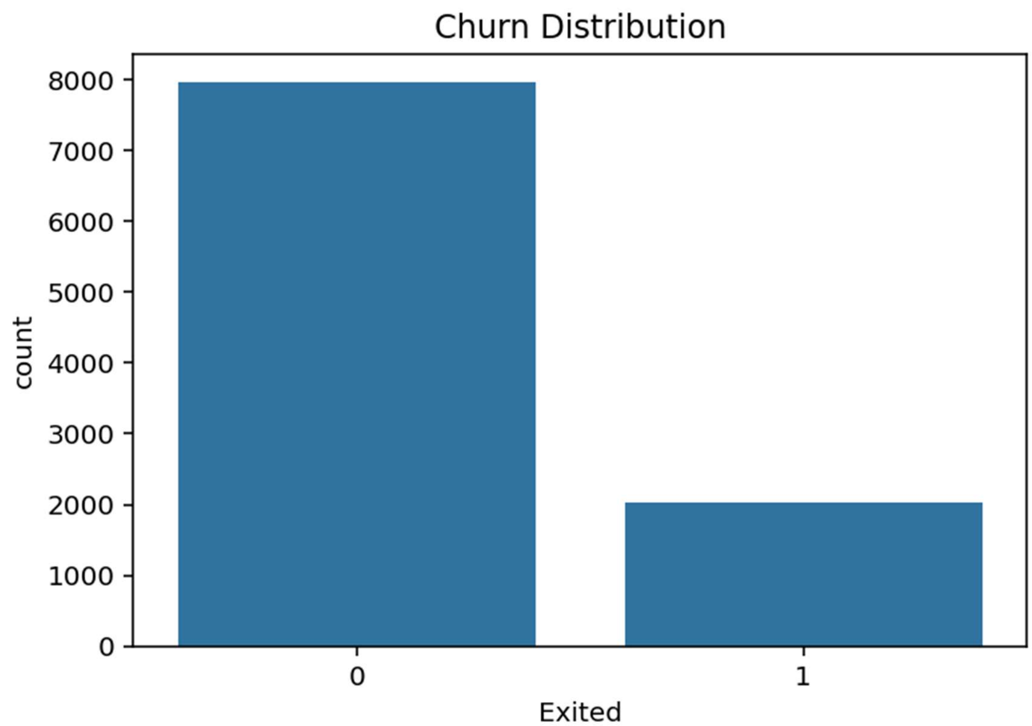

Figure 1. Churn distribution — 20.4% of customers in the dataset have churned.

4. Exploratory Data Analysis

4.1 Distributions of key numeric features

Histograms of the core numeric fields show a fairly typical retail-banking customer base: credit scores cluster around 600-700, ages skew 30s-40s, tenure is evenly spread from 0-10 years, and a large group of customers (roughly a third) carry a zero balance while the rest cluster between 100k-150k.

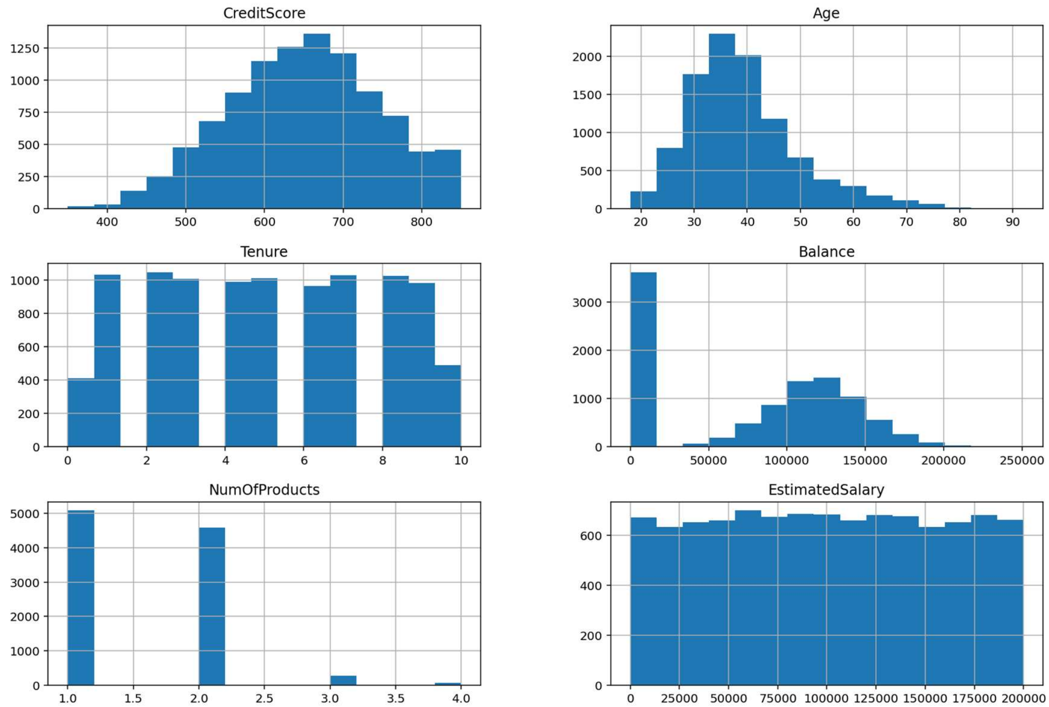

Figure 2. Distribution of CreditScore, Age, Tenure, Balance, NumOfProducts and EstimatedSalary.

4.2 Churn by geography

Germany stands out: it holds fewer total customers than France, but its churn rate is far higher — roughly a third of German customers have left versus about a fifth for France and only around a sixth for Spain. This made Geography (specifically Germany) one of the strongest categorical signals in the dataset.

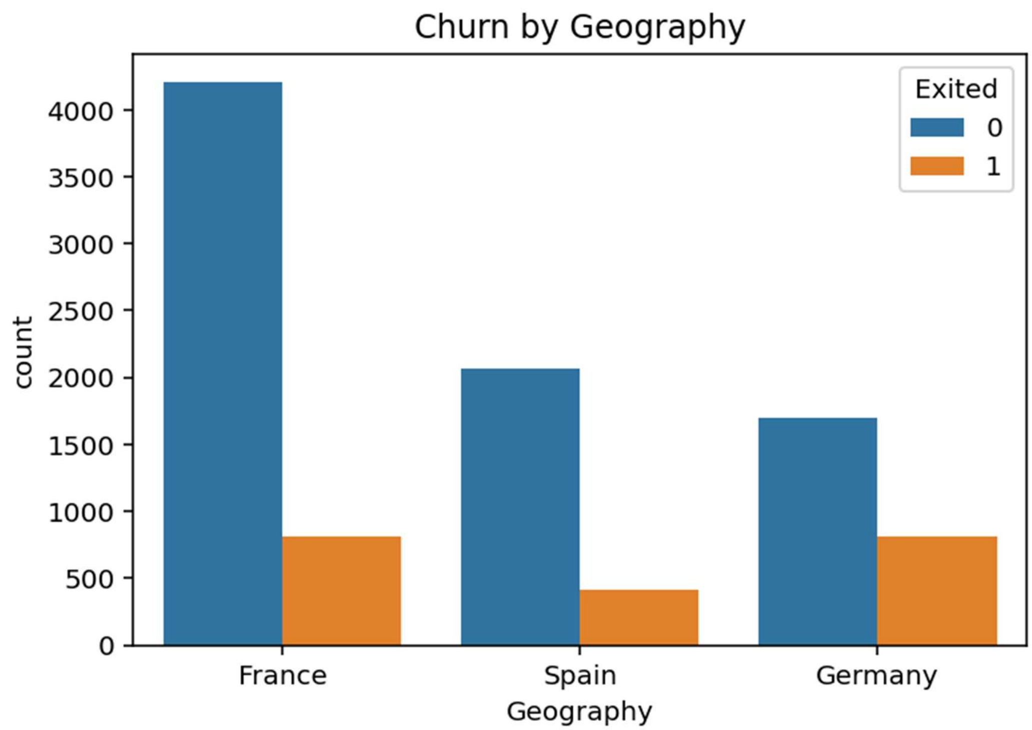

Figure 3. Customer counts by geography, split by churn status.

4.3 Churn by age group

Churn rises steadily with age through the 40s and 50s before tapering off for the 60+ group (a smaller, likely more loyal segment). The 50-60 bracket is the only age band where churned customers actually outnumber retained ones in proportion, making it a natural focus for retention outreach.

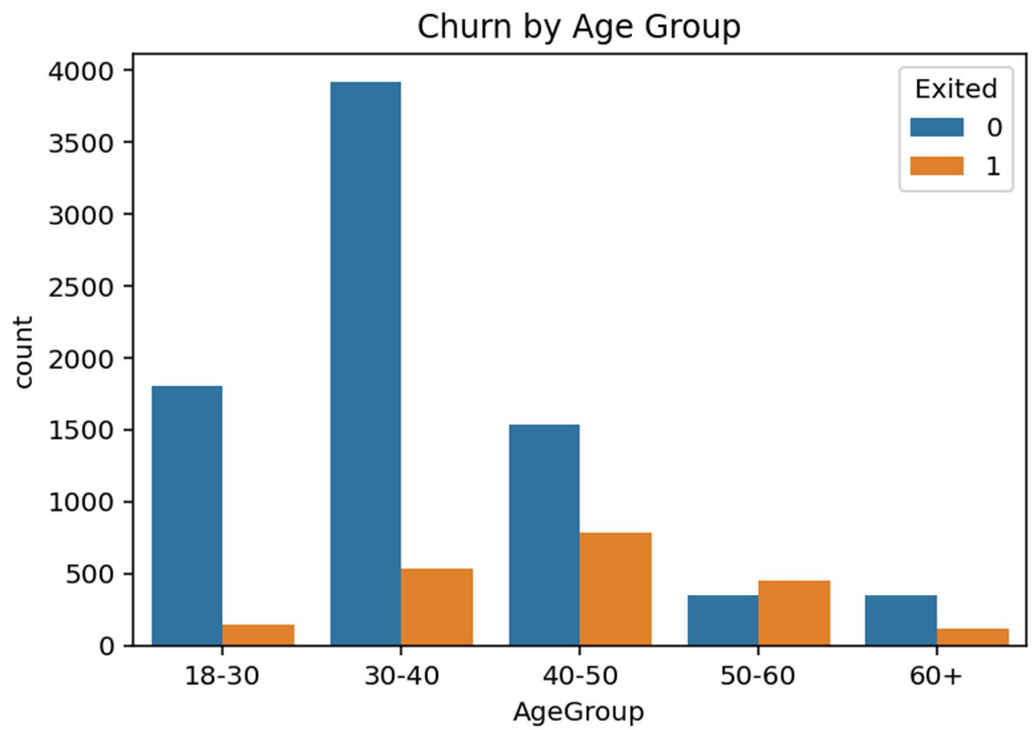

Figure 4. Customer counts by age group, split by churn status.

4.4 Correlation heatmap

Pairwise correlations with Exited are modest in isolation — Age (+0.29) is the strongest single linear correlate, followed by Balance (+0.12) and IsActiveMember (-0.16). This matters: it tells us up front that a purely linear view will miss a lot of the signal, which later shows up as a real gap between the Logistic Regression and tree-based models.

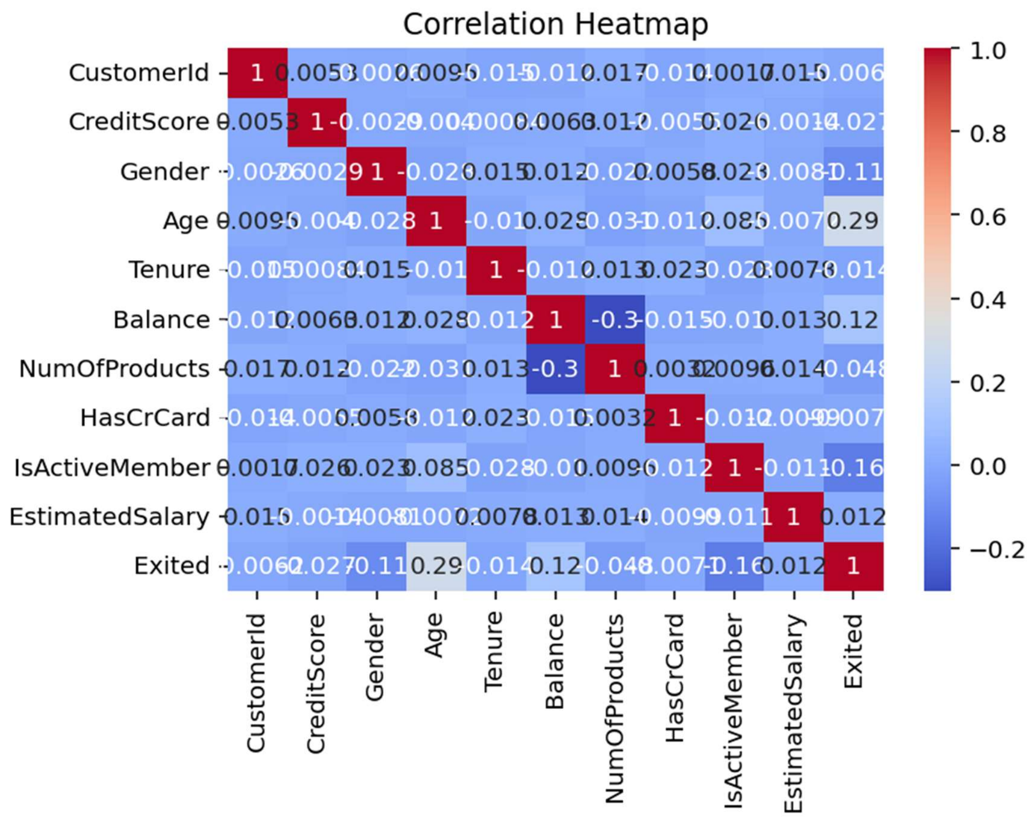

Figure 5. Correlation heatmap across all numeric and encoded features.

4.5 Two-way pattern: geography × activity status

Cutting churn by both geography and IsActiveMember shows the same story in every country: inactive members churn at a noticeably higher rate than active ones. Germany's overall churn rate is elevated even among active members, which is consistent with something structural (pricing, service, or competition) rather than pure disengagement.

5. Feature Importance (first pass)

An initial Random Forest fit — before any final train/test split — was used purely to sanity-check which features looked informative before building the formal modeling pipeline. Age, EstimatedSalary, CreditScore and Balance came out on top, with the one-hot geography flags and age-group dummies contributing comparatively little on their own.

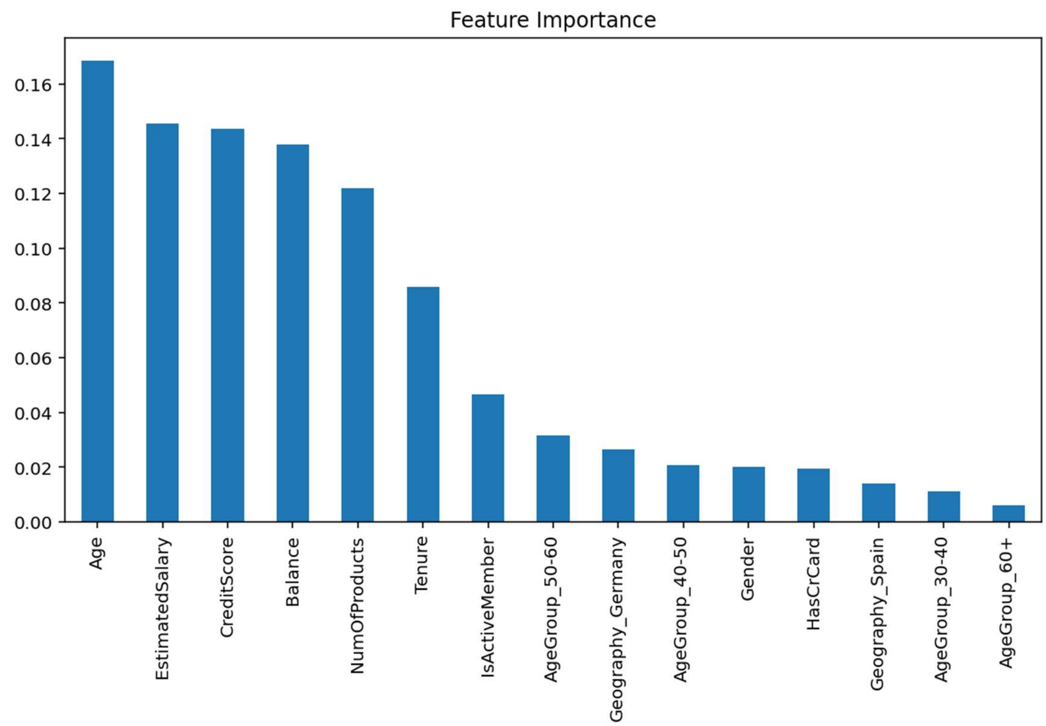

Figure 6. Feature importance from an initial exploratory Random Forest fit.

6. Predictive Modeling

Customers were split 80/20 into train and test sets, stratified on the churn label so both sets kept the same ~20% churn rate. Three classifiers were trained on the test-consistent feature set (CreditScore, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, and one-hot encoded Geography):

- Logistic Regression — class_weight='balanced', max_iter=1000. A linear, highly interpretable baseline.
- Random Forest — 200 trees, class_weight='balanced'. Captures non-linear interactions without much tuning.
- XGBoost — 200 estimators, max_depth=5, learning_rate=0.05. Gradient-boosted trees, generally the strongest of the three off the shelf.

7. Model Evaluation & Comparison

Metrics below were reproduced on a held-out 20% test split (2,000 customers):

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.714	0.387	0.700	0.499	0.777
Random Forest	0.863	0.782	0.450	0.571	0.855
XGBoost	0.867	0.793	0.469	0.590	0.867

Logistic Regression catches the most churners (recall 0.70) but at the cost of a lot of false alarms — its precision of 0.39 means most customers it flags as "at risk" actually stay. Random Forest and XGBoost trade some of that recall for much better precision and a materially higher ROC-AUC, meaning they rank at-risk customers more reliably even though, at a standard 0.5 cutoff, they catch a smaller share of churners outright. XGBoost is the best all-round performer and is the model used downstream for the risk scoring and dashboard.

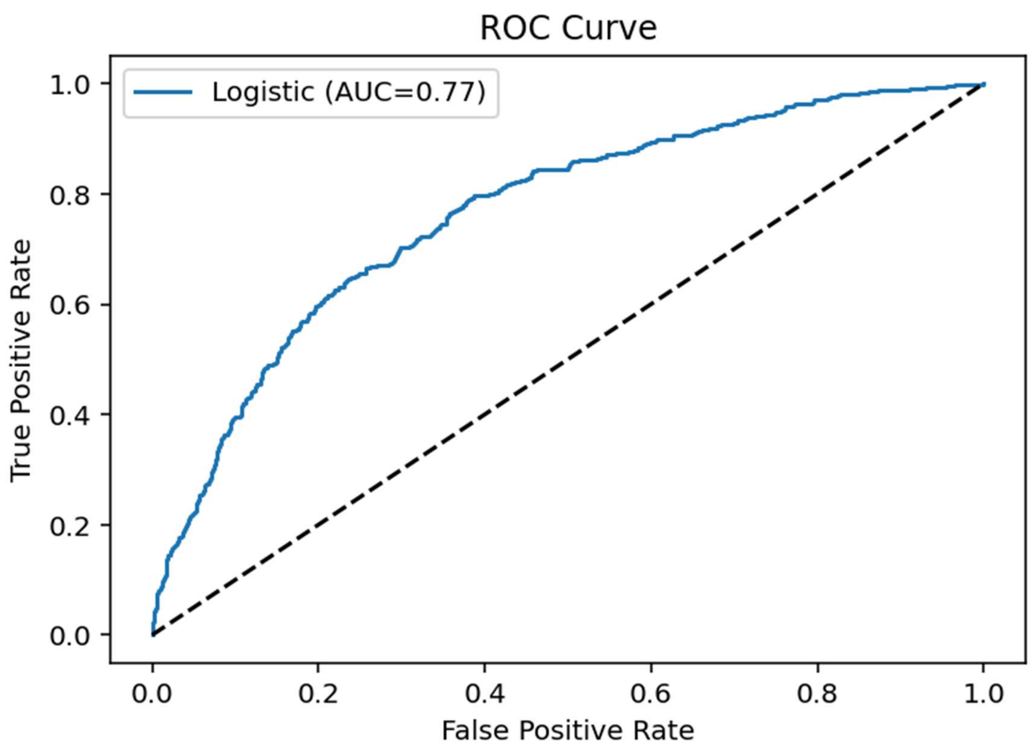

Figure 7. ROC curve for the Logistic Regression baseline (AUC = 0.78).

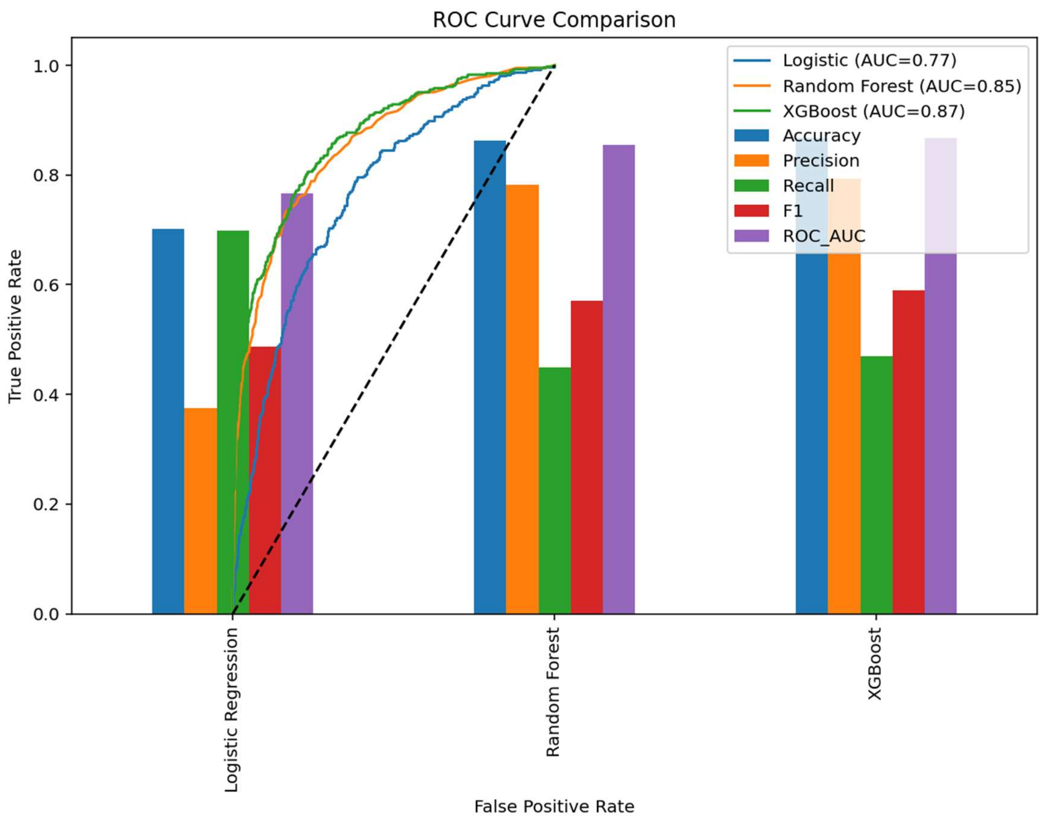

Figure 8. ROC curves for all three models overlaid with their headline metrics.

8. Key Churn Drivers

Ranking XGBoost's feature importances puts NumOfProducts, IsActiveMember, and Age clearly ahead of everything else, with Geography_Germany a distant-but-real fourth. Balance, Gender, and CreditScore contribute modestly; Tenure, HasCrCard and EstimatedSalary barely move the model.

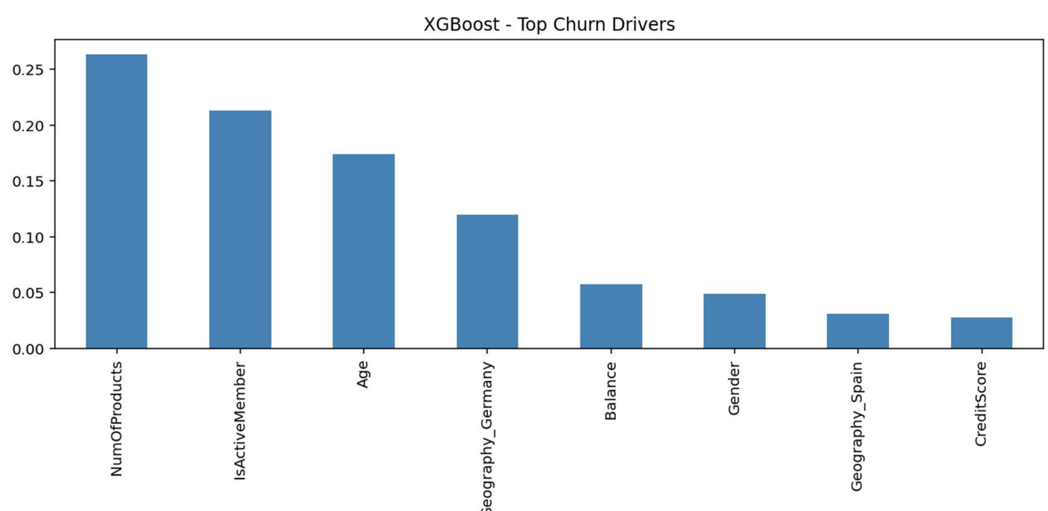

Figure 9. Top churn drivers according to XGBoost's feature importance.

8.1 Why the linear and tree-based views disagree

Logistic Regression's coefficients tell a different story: Geography_Germany and IsActiveMember dominate, with Gender third and NumOfProducts a distant fourth. This isn't a contradiction — it reflects the fact that NumOfProducts has a non-linear, U-shaped relationship with churn (customers with 1 product churn moderately, customers with 2 products churn least, and customers with 3-4 products churn the most), which a linear model compresses into a small average effect while a tree-based model can represent directly by splitting on the value.

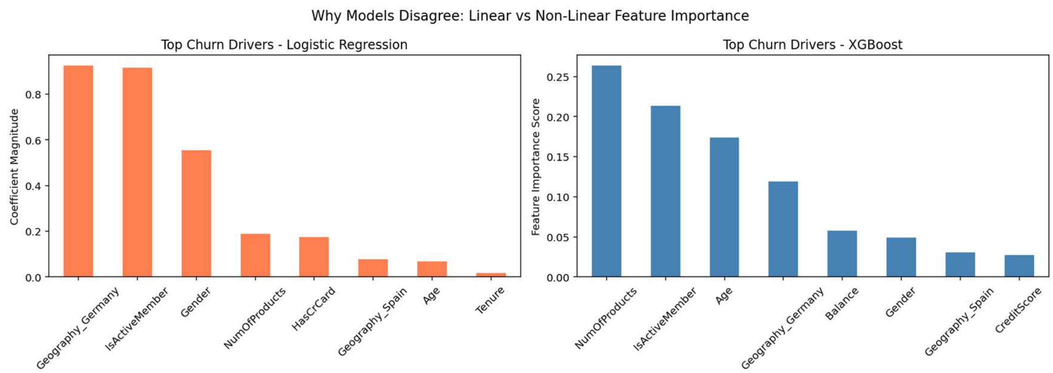

Figure 10. Feature importance side by side — Logistic Regression coefficients (linear) vs XGBoost importances (non-linear).

9. Customer Risk Segmentation

Every test-set customer was scored with XGBoost's predicted churn probability and assigned to a risk tier:

Risk Tier	Probability	Recommended Action Focus
High Risk	> 0.60	Immediate, high-touch retention offer
Medium Risk	0.30 - 0.60	Targeted nudges, re-engagement
Low Risk	< 0.30	Standard loyalty communication

Applied to the 2,000-customer test set, the distribution skewed heavily toward Low Risk, with a smaller High Risk group that retention efforts can realistically focus on first.

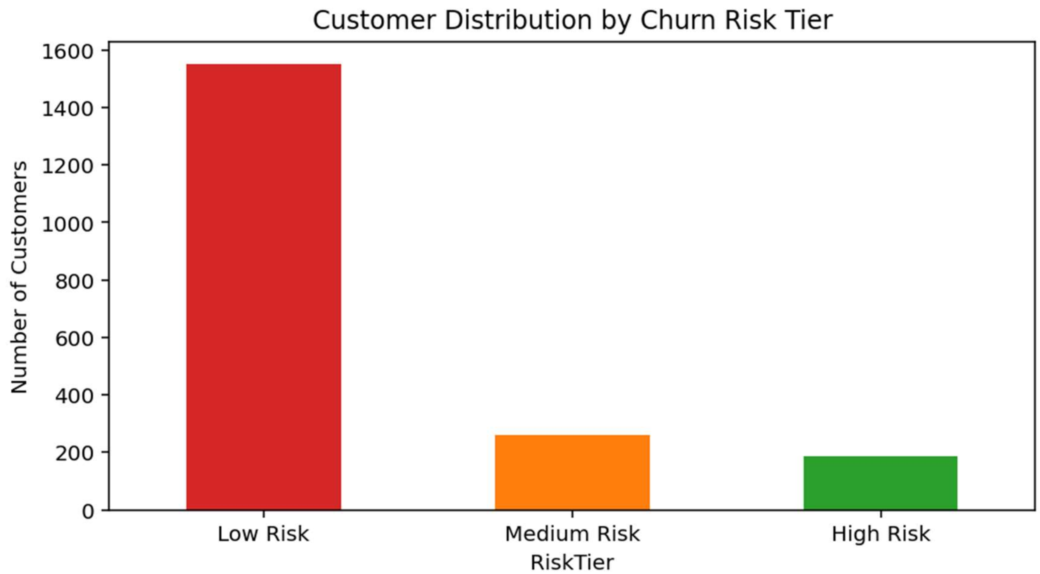

Figure 11. Customer counts by churn risk tier.

10. Retention Strategy by Risk Tier

High Risk

- Assign a dedicated relationship manager
- Offer a personalised retention discount or loyalty reward
- Proactive outreach call within 7 days
- Offer an upgraded product bundle (addresses the NumOfProducts driver)
- Run a re-engagement campaign specifically for inactive members

Medium Risk

- Send a targeted email highlighting unused features or benefits
- Offer a flexible plan adjustment to ease balance-driven financial stress
- Germany-specific: regional trust-building campaigns, given the elevated baseline churn there
- Prompt customers to activate membership benefits (addresses the IsActiveMember driver)

Low Risk

- Standard loyalty-programme communications
- Cross-sell a second product where the customer only holds one
- Annual satisfaction survey to catch early signs of dissatisfaction

11. Limitations & Next Steps

- The dataset is a single snapshot in time — there's no history of behaviour change leading up to churn, which would likely improve early warning.
- Precision at High Risk should be validated against a live holdout before committing budget to retention offers at scale.
- Consider hyperparameter tuning (grid/Bayesian search) and probability calibration for XGBoost before production deployment.
- The interactive dashboard built from this analysis uses a logistic-style scoring function calibrated to this dataset for transparency; the full XGBoost model would need to be served via an API for maximum accuracy in production.

12. Appendix — Full Metric Definitions

Accuracy: share of all predictions that were correct. Precision: of customers predicted to churn, the share that actually did. Recall: of customers who actually churned, the share the model caught. F1: harmonic mean of precision and recall. ROC-AUC: probability the model ranks a random churner above a random non-churner, independent of any specific cutoff threshold.